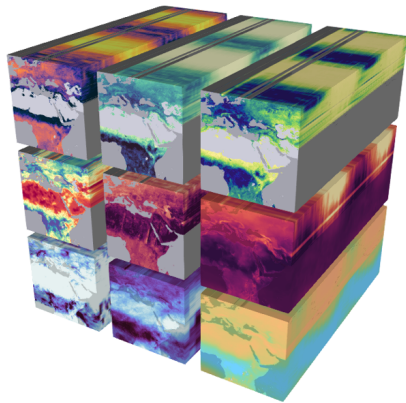


A Friendly Introduction to Dimensionality Reduction

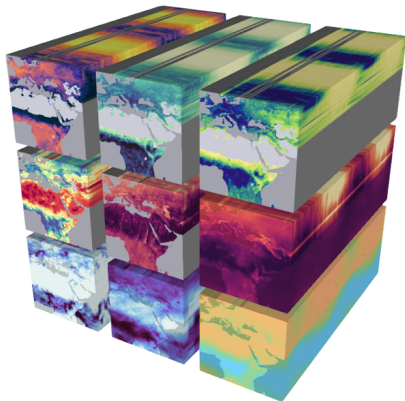
Nathan Mankovich

Earth systems data

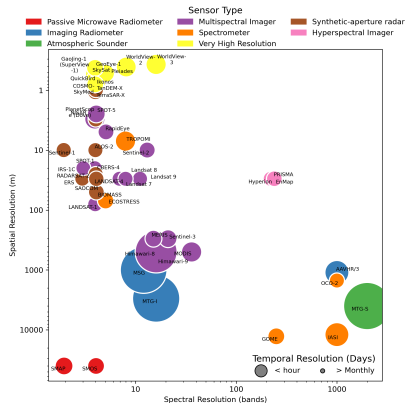


Earth system data cubes Mahecha
et al. [2020].

Earth systems data

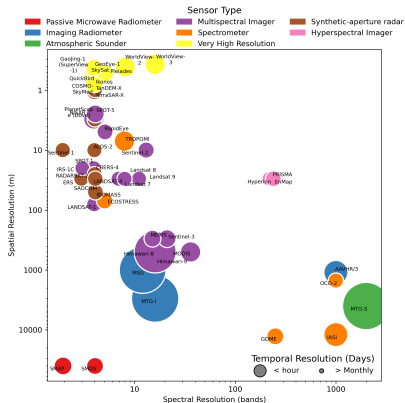


Earth system data cubes Mahecha et al. [2020].

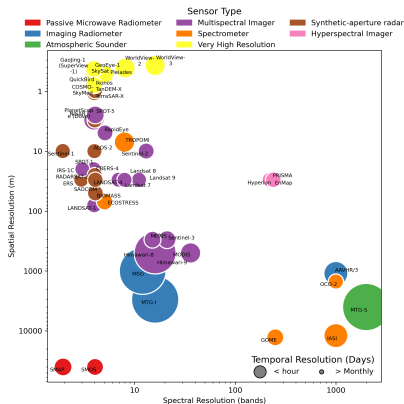


Dimensionality of RS data modalities.

Earth-systems data is high dimensional

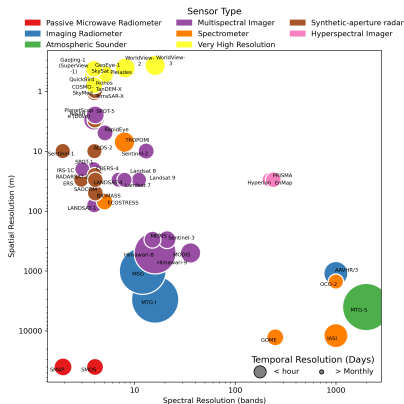


Earth-systems data is high dimensional



- ▶ Feature dimensions: spatial, spectral, temporal
- ▶ Sensors are not often high-dimensional in all dimensions
- ▶ Different data modalities in RS & outside of RS

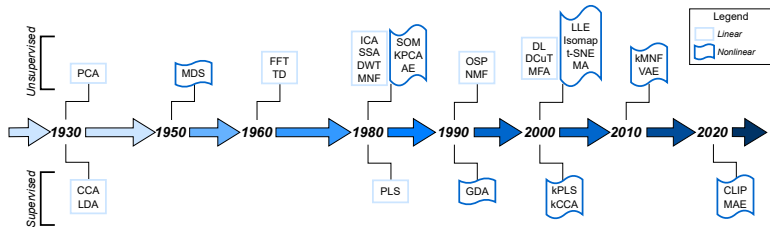
Earth-systems data is high dimensional



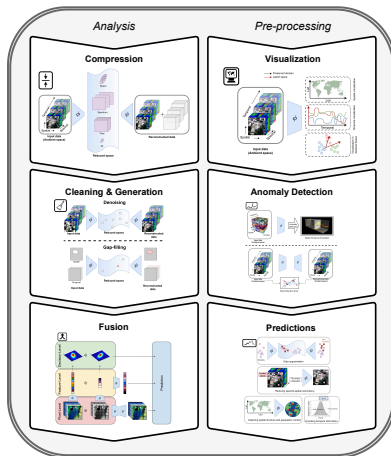
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Dimensionality Reduction (DR) extracts a smaller number of discriminatory features.

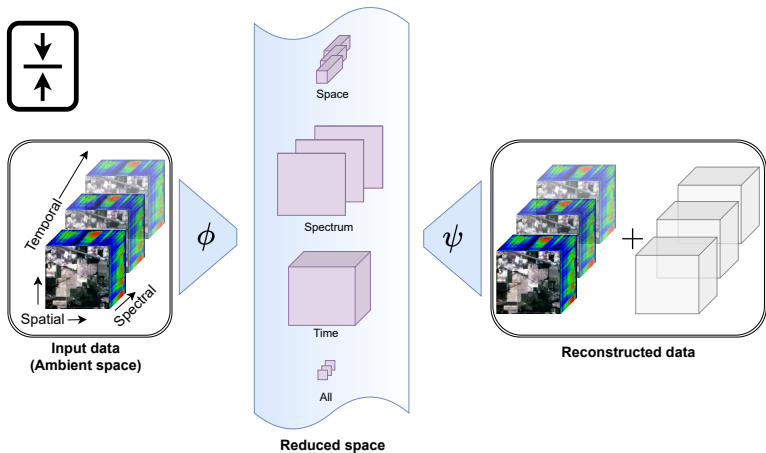
Dimensionality Reduction over the Years



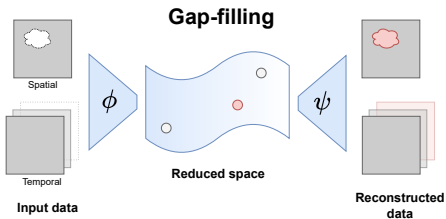
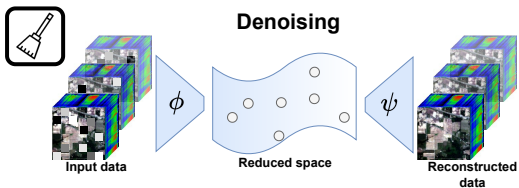
Dimensionality Reduction in Remote Sensing

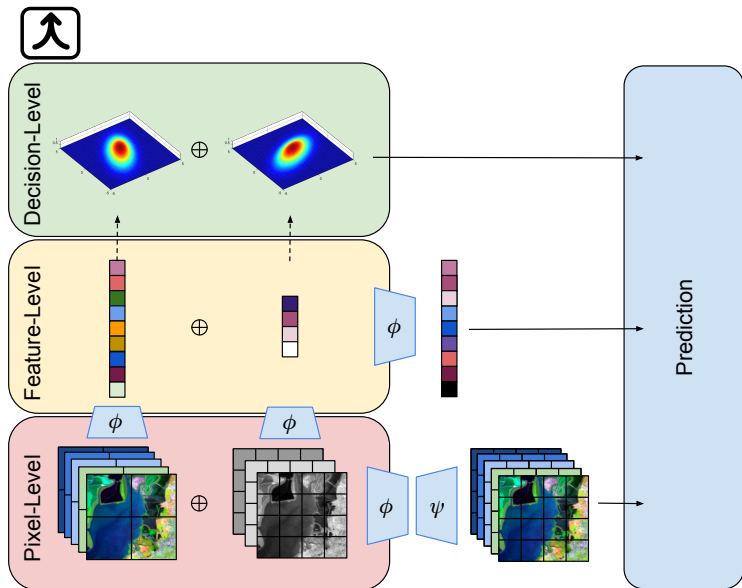


Compression



Denoising

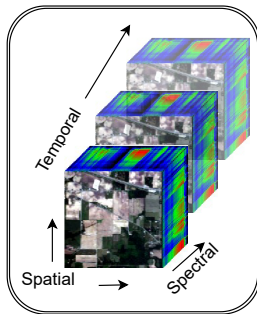




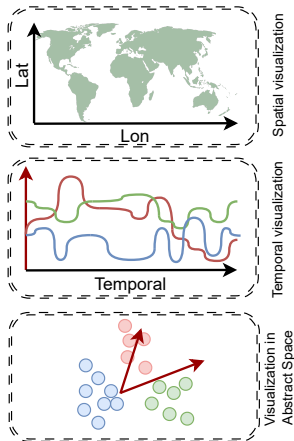
Visualization



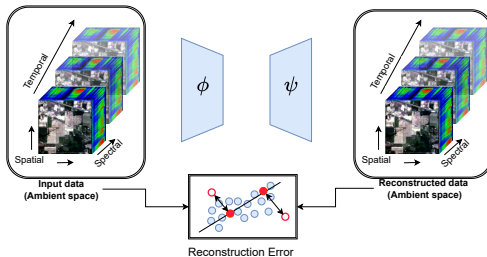
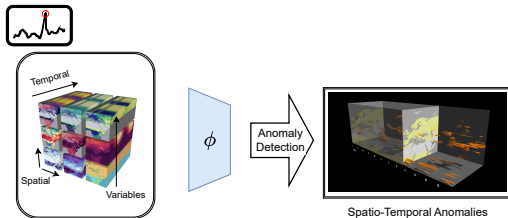
→ Preserved domain
→ Latent space



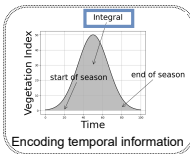
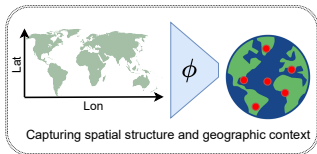
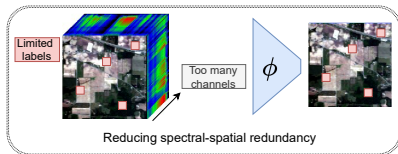
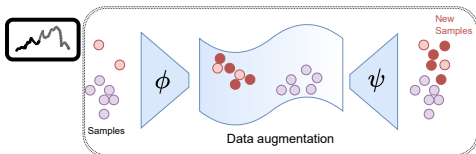
**Input data
(Ambient space)**



Anomaly Detection

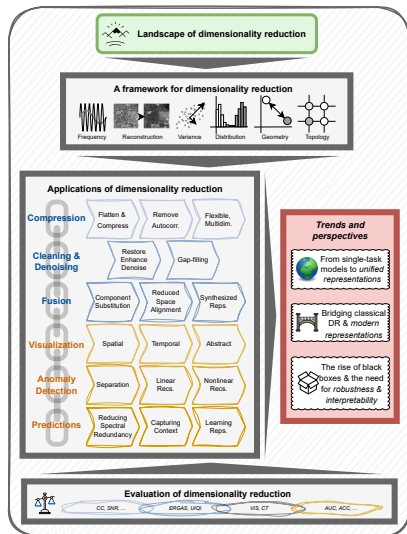


Improved Predictions



Dimensionality Reduction in Remote Sensing (Full Article)

Dimensionality Reduction in Remote Sensing (Full Article)



Dimensionality Reduction for Remote Sensing Data Analysis: A Systematic Review of Methods and Applications

Nathan Mankovich, Kai-Hendrik Cohrs, Homer Durand, Vasileios Sitokoustantinou, Tristan Williams, and Gustau Camps-Valls
Image Processing Lab, Universitat de València, C/ Cat. Agustín Escardino Benlloch, 9, Paterna, 46980, Valencia, Spain

Link to article:

<https://arxiv.org/pdf/2510.18935>

1. Linear Algebra Review
2. Intro to DR
3. Linear Dimensionality Reduction
4. Nonlinear Dimensionality Reduction
5. Autoencoders

Course Webpage

<https://github.com/nmank/DRCourse2025>

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Lab3_PCA.ipynb	Day1	5 months ago
Lab4_LDA.ipynb	Day1	5 months ago
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Lab6_tSNE.ipynb	dr course modified	5 months ago
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A course on dimensionality reduction

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Linear Algebra Review

Further reading [Strang, 2000]

1. Vectors & Inner Products
2. Subspaces, Orthogonality, Projections
3. Matrix Decompositions
4. Gradient Descent

Vectors & Inner Products

Field Axioms

A field F is a set equipped with two operations (addition and multiplication) satisfying the following axioms for all $a, b, c \in F$:

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- ▶ Addition: $a + (b + c) = (a + b) + c$

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- ▶ **Distributivity:** $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

The set of all real numbers (denoted $\mathbb{R}$) is a Field

- ▶ Addition and multiplication operations
- ▶ Additive identity is 0
- ▶ Multiplicative identity is 1

Exercise: Show that $\mathbb{R}$ satisfies the axioms of a field, whereas $\mathbb{Z}$ (the set of all integers) does not.

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- ▶ **Commutativity of addition:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- ▶ **Additive identity:** There exists $\mathbf{0} \in V$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$
- ▶ **Additive inverse:** For each $\mathbf{v} \in V$, there exists $-\mathbf{v} \in V$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
- ▶ **Compatibility with scalar multiplication:** $a(b\mathbf{v}) = (ab)\mathbf{v}$
- ▶ **Identity element of scalar multiplication:** $1\mathbf{v} = \mathbf{v}$, where 1 is the multiplicative identity in F
- ▶ **Distributivity over vector addition:** $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
- ▶ **Distributivity over field addition:** $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

$\mathbb{R}^2$ is a Vector Space

$\mathbb{R}^2$ is a Vector Space over $\mathbb{R}$.

$$\mathbb{R}^2 = \left\{ \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} : a_1, a_2 \in \mathbb{R} \right\}$$

Exercise: Show that $\mathbb{R}^2$ satisfies the axioms of a vector space.

Dot Product & Friends

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► Transpose

$$\mathbf{a}^{\top} = [a_1 \quad a_2]$$

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- ▶ **Angle between vectors**

$$\cos(\theta) = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2}$$

Span

The *span of two vectors* $\mathbf{a}, \mathbf{b}$ is the set of all vectors that can be written as a linear combination of $\mathbf{a}$ and $\mathbf{b}$

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$$\text{span}(\mathbf{a}, \mathbf{b}) = \alpha \mathbf{a} + \beta \mathbf{b} : \alpha, \beta \in \mathbb{R}$$

Linear Independence

A set of vectors $\{v_1, v_2, \dots, v_k\}$ in a vector space V is **linearly independent** if:

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k = 0 \quad \Rightarrow \quad c_1 = c_2 = \dots = c_k = 0$$

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In other words, the only solution to the linear combination equaling zero is the **trivial solution**.

Example:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

are linearly independent in $\mathbb{R}^2$, since neither can be written as a multiple of the other.

What is a Basis?

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A **basis** of a vector space V is a set of vectors $\{v_1, v_2, \dots, v_n\} \subset V$ such that:

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Example: The standard basis for $\mathbb{R}^3$ is:

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The **dimension** of a vector space V is the number of elements in a basis for V .

Subspaces, Orthogonality, Projections

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

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- ▶ The **row space** of $\mathbf{A}$ (denoted $\text{row}(\mathbf{A})$) is the set of linear combinations of rows.

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- ▶ The **column space** of $\mathbf{A}$ (denoted $\text{col}(\mathbf{A})$) is the set of linear combinations of columns.

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- ▶ The **trace** of $\mathbf{A}$ is the sum of the diagonal entries

$$\text{tr}(\mathbf{A}) = a_{11} + a_{22}$$

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- ▶ The Frobenius norm of $\mathbf{A}$ is

$$\|\mathbf{A}\|_F = \sqrt{\text{tr}(\mathbf{A}^\top \mathbf{A})}$$

Matrix multiplication

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

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Projections

Projections

Projection of $\mathbf{b}$ onto $\text{col}(\mathbf{A})$:

$$\Pi_{\mathbf{A}}(\mathbf{b}) = \mathbf{A}(\mathbf{A}^{\top}\mathbf{A})^{-1}\mathbf{A}^{\top}\mathbf{b}$$

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Exercise: If the columns of $\mathbf{A}$ are orthogonal, show that the projection onto the column space of $\mathbf{A}$ is $\mathbf{A}\mathbf{A}^{\top}$.

Lab 1: Vectors & Matrices

Go to `Lab1_VectorsMatrices.ipynb`

Matrix Decompositions

Singular Value Decomposition (SVD)

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- ▶ $\mathbf{U} \in \mathbb{R}^{p \times r}$: left singular vectors, $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$
- ▶ $\mathbf{\Sigma} \in \mathbb{R}^{r \times r}$: diagonal matrix of singular values
- ▶ $\mathbf{V}^\top \in \mathbb{R}^{r \times d}$: right singular vectors, $\mathbf{V}^\top \mathbf{V} = \mathbf{I}$

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- ▶ $\mathbf{U} \in \mathbb{R}^{p \times r}$: left singular vectors, $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$
- ▶ $\mathbf{\Sigma} \in \mathbb{R}^{r \times r}$: diagonal matrix of singular values
- ▶ $\mathbf{V}^\top \in \mathbb{R}^{r \times d}$: right singular vectors, $\mathbf{V}^\top \mathbf{V} = \mathbf{I}$

Applications: dimensionality reduction, image compression, linear systems.

Eigenvalue Decomposition from SVD

Given the SVD

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Exercises:

- ▶ Show the eigenvalue decomposition of $\mathbf{A}\mathbf{A}^\top$ is $\mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^\top \mathbf{U}^\top$
- ▶ Show the trace of $\mathbf{C}$ is the sum of its eigenvalues

Eigenvalue Optimization Formulation

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1. Rayleigh quotient:

$$\lambda_i = \max_{\mathbf{v}^\top \mathbf{v}_j = 0 \ \forall j < i} \frac{\mathbf{v}^\top \mathbf{C} \mathbf{v}}{\mathbf{v}^\top \mathbf{v}}$$

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Exercise: Show that these optimizations are equivalent.

Generalized Eigenvalue Decomposition

Solve:

$$\mathbf{C}_A \mathbf{w} = \lambda \mathbf{C}_B \mathbf{w}$$

Optimization form:

$$\lambda_i = \max_{\mathbf{w}^\top \mathbf{w}_j = 0 \ \forall j < i} \frac{\mathbf{w}^\top \mathbf{C}_A \mathbf{w}}{\mathbf{w}^\top \mathbf{C}_B \mathbf{w}}$$

Approximate with:

$$\mathbf{C}_B^{-1} \mathbf{C}_A$$

Use SVD for pseudo-inverse:

$$\blacktriangleright \mathbf{C}_B = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$$

$$\blacktriangleright \mathbf{C}_B^\dagger = \mathbf{V} \mathbf{\Sigma}^{-1} \mathbf{U}^\top$$

Condition number:

$$\kappa(\mathbf{C}_B) = \frac{\sigma_{\max}}{\sigma_{\min}}$$

Big κ bad, Small κ good

Lab 2: Matrix Decompositions

Go to `Lab2_MatrixDecompositions.ipynb`

The Landscape of Dimensionality Reduction

Inspired by Lee and Verleysen [2007]

DR Reduces the Feature Space

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Dimensionality Reduction (DR) maps d -dimensional features to $k < d$ -dimensional “essential” features

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- ▶ p samples
- ▶ d dimensional ambient space
- ▶ Input dataset $\mathcal{D}$ (d features)
- ▶ Output: reduced dataset $\mathcal{R}$ ($k < d$ features)

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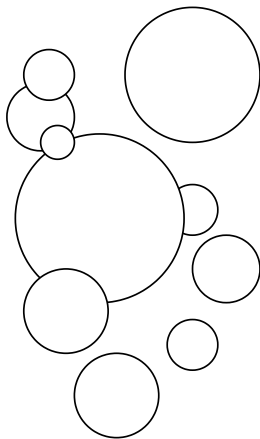
- ▶ p samples
- ▶ d dimensional ambient space
- ▶ Input dataset $\mathcal{D}$ (d features)
- ▶ Output: reduced dataset $\mathcal{R}$ ($k < d$ features)

Many DR algorithms & definitions of “essential”

Toy Example: Circles

Toy Example: Circles

Ambient space



Reduced space

$(0, 0), 1$

$(2, 3), 2$

$(-1, 4), 1.5$

$(3, -2), 0.8$

$(-2, -3), 2.5$

$(1, -1), 1.2$

$(-3, 1), 0.9$

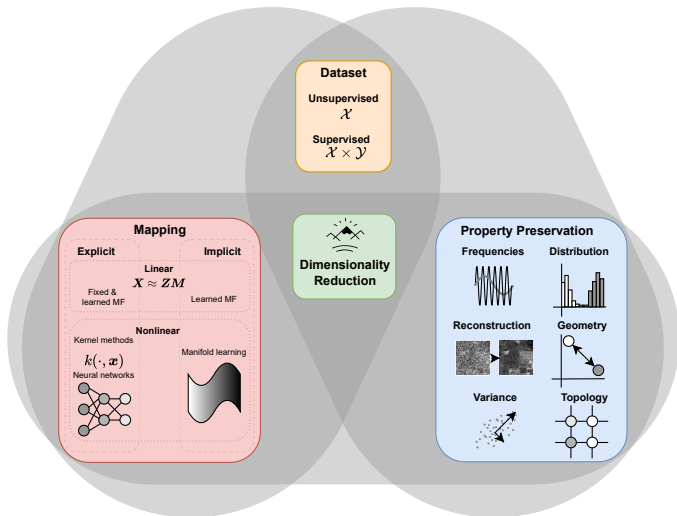
$(4, 4), 1.8$

$(-4, -2), 1.1$

$(0, 5), 2$

DR Summary- Properties

DR Summary- Properties



Unsupervised

$$\mathcal{D} = \mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p\} \subset \mathbb{R}^d$$

e.g., Visualize data in 2D to see if there are any patterns

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Supervised

$$\mathcal{D} = \{\mathcal{X}, \mathcal{Y}\}, = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p\}, \{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_p\}$$

e.g., Find a low-dimensional, discriminatory feature space w.r.t. $\mathcal{Y}$

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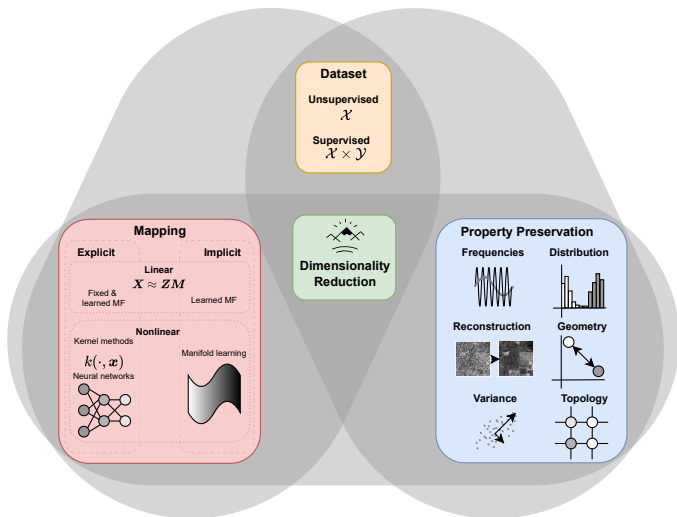
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e.g., Find a low-dimensional, discriminatory feature space w.r.t. $\mathcal{Y}$

Semi-supervised & self-supervised methods break this paradigm... but we will not discuss these methods in this course

DR Summary- Properties

DR Summary- Properties



Explicit DR mapping outputs ϕ
where

$$\phi(\mathbf{x}_i) = \mathbf{z}_i$$

► Approximate inverse
 $\psi \approx \phi^{-1}$

► Reconstructions:
 $\mathbf{x}_i \approx \hat{\mathbf{x}}_i = \psi(\phi(\mathbf{x}_i))$

e.g., Want to fit a model on
some data, then apply it to
“unseen” data.

Implicit: ϕ, ψ are not explicitly
defined but inferred

e.g., Model fit on all the data to
be reduced

Matrix Factorization

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Data matrix: $\mathbf{X} = [\mathbf{x}_1 \cdots \mathbf{x}_p]^\top \in \mathbb{R}^{p \times d}$

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The model:

$$\mathbf{X} \approx \mathbf{Z}\Psi$$

Example: principal component analysis factors data matrix $\mathbf{X}$ into

- ▶ Principal components (rows of $\mathbf{Z}$)
- ▶ Empirical orthogonal functions (columns of Ψ)

Kernel Methods: Nonlinear Generalizations of Matrix Factorization

- ▶ Classical DR: **explicit linear mapping**

$$\mathbf{z} = \mathbf{\Psi}^\top \mathbf{x}$$

- ▶ Kernel methods extend classical DR to **nonlinear setting** by mapping into a high-dimensional feature space

$$\phi : \mathbf{x} \mapsto \mathcal{H},$$

where linear DR is performed.

- ▶ **Kernel trick** replaces explicit feature mappings with pairwise similarities

$$k(\mathbf{x}_i, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle,$$

enabling nonlinear embeddings without constructing ϕ .

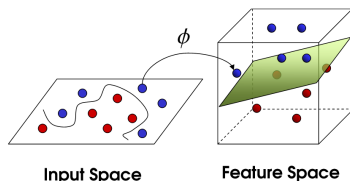


Figure: Image from <https://medium.com/data-science/the-kernel-trick-c98cdbcaeb3f>

Neural Networks

- ▶ Neural network-based DR methods learn a **parametric, nonlinear mapping**

$$f_{\theta} : \mathbf{x} \mapsto \mathbf{z},$$

- ▶ f_{θ} defined by a composition of affine transformations and nonlinear activations.
- ▶ Like kernel methods, mapping is learned **explicitly**.
- ▶ Objective function defines DR properties (discussed later)

Example: Autoencoder

- ▶ Encoder: $\phi(\mathbf{x}) = \mathbf{z}$,
Decoder: $\psi(\mathbf{z}) = \hat{\mathbf{x}}$
- ▶ Loss: e.g., MSE
 $\sum_i \|\mathbf{x}_i - \psi \circ \phi(\mathbf{x}_i)\|^2$

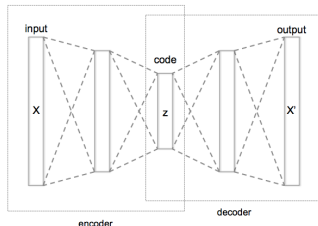


Figure: Image from https://commons.wikimedia.org/wiki/File:Autoencoder_structure.png

Manifold Learning: Implicit Mapping via Data Geometry

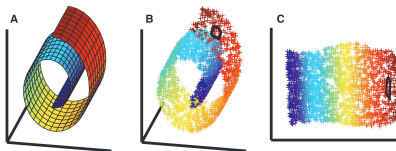
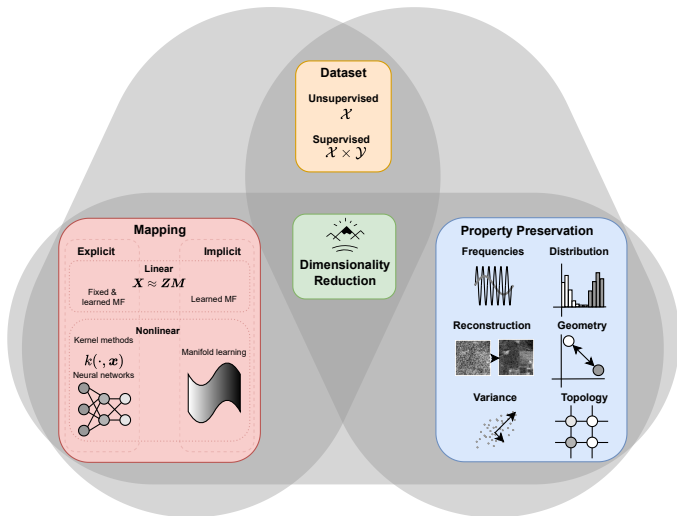


Figure: Image from <https://www.dissonances.blog/p/ambient-spaces>.

- ▶ Assume high-dimensional observations lie on or near a low-dimensional **manifold** embedded in the ambient space.
- ▶ **Implicit** mappings to low-dimensional representations by preserving local geometric or topological relationships.
- ▶ The embedding is typically obtained by solving a graph-based or spectral problem derived from neighborhood relations in the data.
- ▶ The resulting coordinates describe the intrinsic manifold structure.

DR Summary- Properties

DR Summary- Properties



DR as Optimization

DR reduced space representation $\mathcal{Z}$ is found as minimizer of a loss function L of the data $\mathcal{X}$.

$$\begin{aligned} \arg \min_{\mathcal{Z}} \quad & L(\mathcal{X}, \mathcal{Z}) \\ \text{s.t.} \quad & \Omega(\mathcal{Z}, \mathcal{X}) = 0 \end{aligned}$$

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- ▶ Distributions
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but not all properties can be preserved at once

Frequency and signal structure preserving

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- ▶ Exploit **pre-defined basis** or transform aligned with known signal structures in the data (e.g., frequencies, scales).

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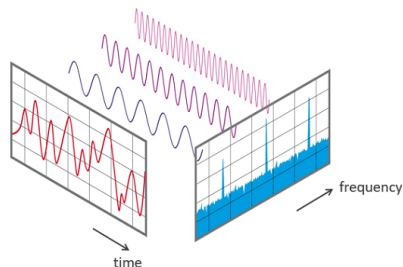


Figure: Image from <https://www.nti-audio.com/en/support/know-how/fast-fourier-transform-fft>

Reconstruction-preserving

For each $i = 1, 2, \dots, p$ we have

$$\mathbf{x}_i \approx \psi \circ \phi(\mathbf{x}_i)$$

Loss function

$$\begin{aligned} \min_{\psi, \phi} \quad & \sum_{i=1}^p \|\mathbf{x}_i - \psi \circ \phi(\mathbf{x}_i)\|^2 \\ \text{s.t.} \quad & \phi : \mathbb{R}^d \rightarrow \mathbb{R}^k, \quad \psi : \mathbb{R}^k \rightarrow \mathbb{R}^d \end{aligned} \tag{1}$$

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Useful when interpretability or invertibility is important. Tasks like compression, denoising, inpainting, generative modeling.

Variance-preserving

Data $\mathbf{x}$ and DR mapping $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^k$

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Example: Principal component analysis Hotelling [1933].

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Interpretable and useful for detecting modes or variation, noise reduction, compression, and exploratory analysis.

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Useful for detecting clusters, visualization, generative modeling, and anomaly detection.

Non-linear DR

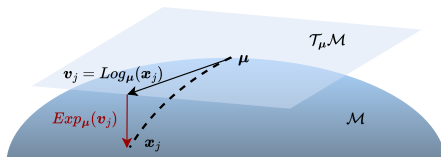
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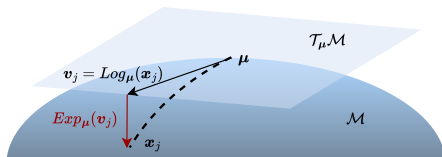
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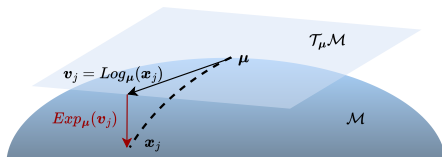
Local Linearity: Around each $\mathbf{x}_i \in \mathcal{M}$, there exists a neighborhood U such that:

$$\mathcal{M} \cap U \approx T_{\mathbf{x}_i} \mathcal{M}$$

where $T_{\mathbf{x}_i} \mathcal{M}$ is the tangent space at $\mathbf{x}_i$.

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where $T_{\mathbf{x}_i} \mathcal{M}$ is the tangent space at $\mathbf{x}_i$.

Geodesic Distances: For $\mathbf{x}_i, \mathbf{x}_j \in \mathcal{M}$, define

$$d_{\mathcal{M}}(\mathbf{x}_i, \mathbf{x}_j) = \text{length of shortest path on } \mathcal{M}$$

- ▶ Focuses on preserving distances or neighborhood relationships between points when reducing from $\mathbb{R}^d$ to $\mathbb{R}^k$.

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- ▶ Useful for understanding nonlinear structure, visualization, and discovering clusters or smooth transitions.
- ▶ Example: multidimensional scaling.
- ▶ Some approaches work directly from distance matrices rather than raw feature vectors.

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Data-driven topology (learned):

- ▶ Construct graph $G = (\mathcal{X}, E)$ from data.
- ▶ Edge weights: $w_{ij} = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{\sigma^2}\right)$ or k -nearest neighbors.
- ▶ Preserve graph in low-dimensional representation

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Data-driven topology (learned):

- ▶ Construct graph $G = (\mathcal{X}, E)$ from data.
- ▶ Edge weights: $w_{ij} = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{\sigma^2}\right)$ or k -nearest neighbors.
- ▶ Preserve graph in low-dimensional representation

Predefined topology (fixed):

- ▶ Impose grid or lattice structure: $\mathcal{G} = 1\text{D or } 2\text{D lattice}$.
- ▶ Each reduce-space representation $\mathbf{z}_i$ corresponds to a fixed node in $\mathcal{G}$.

Example: ISOMAP & t-distributed stochastic neighborhood embedding

Comparison:

Geometry-preserving $\Rightarrow$ maintain distances;

Topology-preserving $\Rightarrow$ maintain who is connected to whom.

Too Many DR Methods..

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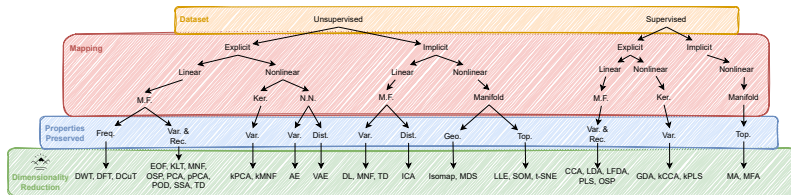
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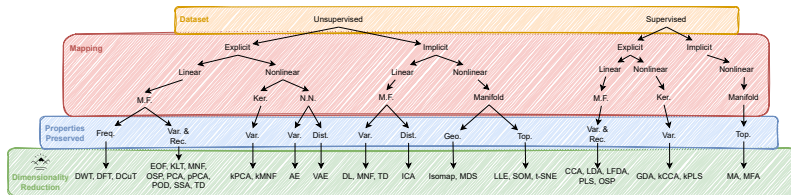
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Useful git repository with a collection of even more methods: <https://github.com/koulakis/awesome-dimensionality-reduction>

Linear Dimensionality Reduction

1. Principal Component Analysis [Hotelling, 1933; Shlens, 2014]
2. Linear Discriminant Analysis [Tharwat et al., 2017]

Principal Component Analysis

We consider a dataset of p samples with d features:

$$\{\mathbf{x}_i\}_{i=1}^p \subset \mathbb{R}^d$$

We collect the dataset into a samples $\times$ features data matrix:

$$\mathbf{X} \in \mathbb{R}^{p \times d}$$

Important: The data must be **mean-centered**:

$$\frac{1}{p} \sum_{i=1}^p \mathbf{x}_i = \mathbf{0}$$

PCA Objective Summary

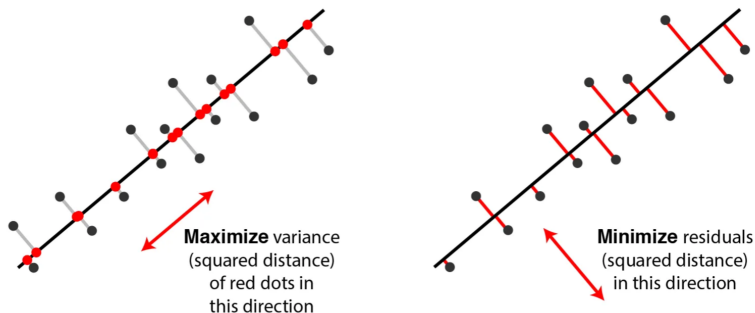


Figure: Image borrowed from <https://medium.com/@fraidoonomarzai99/principal-component-analysis-pca-in-depth-93c871f25dfa>.

Optimization Goal

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$$\mathbf{v}_j = \arg \max_{\substack{\mathbf{v}^\top \mathbf{v} = 1 \\ \mathbf{v}^\top \mathbf{v}_\ell = 0 \ \forall \ell < j}} \text{Var}(\mathbf{X}\mathbf{w}) \quad (3)$$

$$= \arg \max_{\substack{\mathbf{w}^\top \mathbf{w} = 1 \\ \mathbf{w}^\top \mathbf{v}_\ell = 0 \ \forall \ell < j}} \sum_i (\mathbf{x}_i^\top \mathbf{w})^\top (\mathbf{x}_i^\top \mathbf{w}) \quad (4)$$

$$= \arg \max_{\substack{\mathbf{w}^\top \mathbf{w} = 1 \\ \mathbf{w}^\top \mathbf{v}_\ell = 0 \ \forall \ell < j}} \sum_i \|\mathbf{x}_i\|_2^2 \cos \theta(\mathbf{w}, \mathbf{x}_i) \quad (5)$$

$$= \arg \max_{\substack{\mathbf{w}^\top \mathbf{w} = 1 \\ \mathbf{w}^\top \mathbf{v}_\ell = 0 \ \forall \ell < j}} \mathbf{w}^\top \mathbf{X}^\top \mathbf{X} \mathbf{w} \quad (6)$$

Alternative Formulation: Reconstruction Error

This is equivalent to finding the rank- k projection:

$$\Pi_{\mathbf{W}}(\mathbf{x}) := \mathbf{W}\mathbf{W}^{\top}\mathbf{x}$$

We solve:

$$\mathbf{V} = \arg \min_{\mathbf{W}^{\top}\mathbf{W}=\mathbf{I}} \mathbb{E} [\|\mathbf{x}_i - \mathbf{W}\mathbf{W}^{\top}\mathbf{x}_i\|_2^2]$$

Exercise Show that we can find the EOFs (columns of $\mathbf{V}$) via eigendecomposition of $\mathbf{X}^{\top}\mathbf{X}$ or the SVD of $\mathbf{X}$.

Transformation

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PCA reconstruction of an individual sample:

$$\hat{\mathbf{x}} = \psi \circ \phi(\mathbf{x}) = \mathbf{VV}^\top \mathbf{x} = \mathbf{\Pi_V}(\mathbf{x}) \in \mathbb{R}^d$$

Explained Variance

The **explained variance** is how much variance each principal component captures:

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PCA Summary

PCA Summary

Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.		M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn

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- ▶ Have unlabeled data

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Beyond PCA... robust subspace recovery [Lerman and Maunu, 2018] & flag manifolds [Mankovich et al., 2024; Szwagier and Pennec, 2024, 2025]

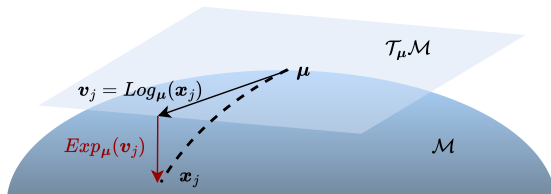
Aside- PCA on Manifolds (Tangent-PCA)

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Assume data samples from known, Riemannian manifold...

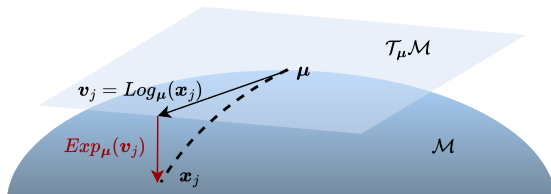
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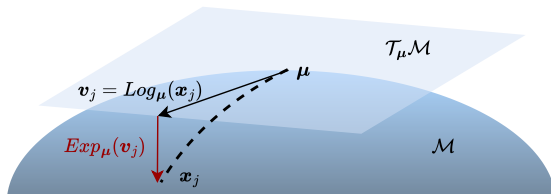
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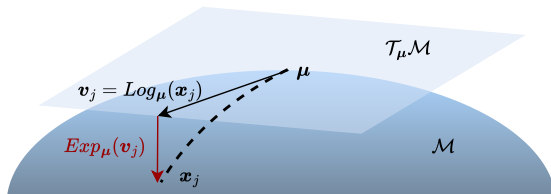
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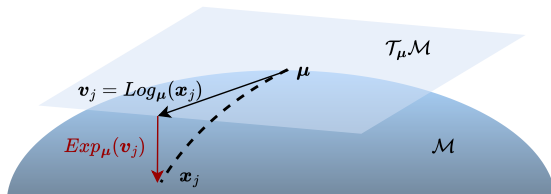
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Read more here: [Fletcher et al., 2004; Pennec, 2018]

Lab 3: PCA

Go to `Lab3_PCA.ipynb`

Linear Discriminant Analysis

What is LDA?

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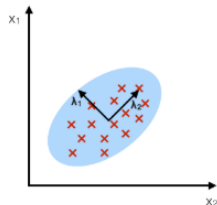
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"PCA with class information!"

LDA Intuition

PCA:

component axes that maximize the variance



LDA:

maximizing the component axes for class-separation

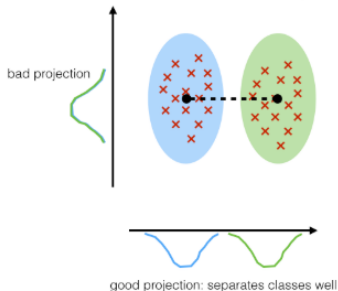


Figure: Image borrowed from
https://sebastianraschka.com/Articles/2014_python_lda.html

Mathematical Formulation

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$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^P, \quad \mathbf{x}_i \in \mathbb{R}^d$$

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Within-class scatter matrix:

$$\mathbf{S}_W = \sum_{c=1}^C \sum_{\mathbf{x}_i \in c} (\mathbf{x}_i - \mu_c)(\mathbf{x}_i - \mu_c)^\top$$

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Optimization Problem

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LDA vs PCA

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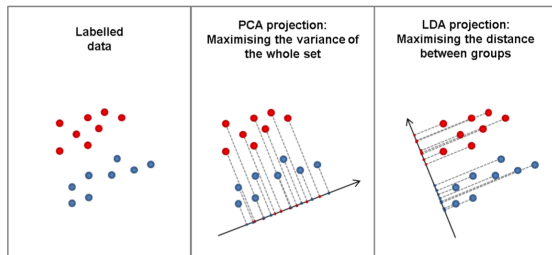


Figure: Image borrowed from <https://vivekmuraleedharan73.medium.com/what-is-linear-discriminant-analysis-lda-7e33ff59020a>.

Aspect	PCA [Shlens, 2014]	LDA [Tharwat et al., 2017]
Supervised?	No	Yes
Objective	Maximize variance	Maximize class separation
Axes chosen	max variance	best separation
Max dimensions	$\leq$ input dimension	$\leq$ number of classes $- 1$

LDA Summary

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- ▶ Labeled data
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- ▶ Want a fast, interpretable linear projection
- ▶ Linearly separable classes

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When to Use LDA

- ▶ Labeled data
- ▶ Want low-dimensional features that separate classes well
- ▶ Want a fast, interpretable linear projection
- ▶ Linearly separable classes
- ▶ Each class follows a multivariate normal distribution with equal covariances

Lab 3: LDA

Go to Lab4_LDA.ipynb

Nonlinear Dimensionality Reduction

1. Multidimensional Scaling (MDS) [Kruskal and Wish, 1978]
2. Isometric Mapping (Isomap) [Balasubramanian and Schwartz, 2002]
3. t-distributed Stochastic Neighborhood Embedding (t-SNE) [Van der Maaten and Hinton, 2008]
4. Uniform Manifold Approximation (UMAP) [McInnes et al., 2018]

Multidimensional Scaling

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Given:

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Goal: Find latent coordinates $\{\mathbf{z}_i\}_{i=1}^p \subset \mathbb{R}^k$ such that

$$\|\mathbf{z}_i - \mathbf{z}_j\|_2 \approx d(x_i, x_j)$$

Useful when there's a non-Euclidean distance between points in the ambient space.

Classical MDS Objective

We solve:

$$\{\mathbf{z}_i\}_{i=1}^p = \arg \min_{\{\mathbf{w}_i\} \subset \mathbb{R}^k} \sum_{i,j} (\|\mathbf{w}_i - \mathbf{w}_j\|_2 - d_{ij})^2$$

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How do we mean-center a distance matrix?

Eigenvalue Decomposition

Let $\mathbf{D}^{(2)}$ be the matrix of squared distances: d_{ij}^2 .

Mean-centering matrix:

$$\mathbf{J} = \mathbf{I} - \frac{1}{p} \mathbf{1}\mathbf{1}^\top$$

Double-mean center matrix of squared distances:

$$\mathbf{B} = -\frac{1}{2} \mathbf{J} \mathbf{D}^{(2)} \mathbf{J}$$

Compute eigenvalue decomposition:

$$\mathbf{B}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}$$

Low-Dimensional Embedding

Truncate to top- k eigenvalues and eigenvectors:

$$\bar{\Lambda}, \bar{\mathbf{V}}$$

Construct the embedding:

$$\mathbf{Z} = \bar{\mathbf{V}} \bar{\Lambda}^{1/2}$$

$\mathbf{Z}^\top = [\mathbf{z}_1, \dots, \mathbf{z}_n]$ are the low-dimensional coordinates.

Note: When $\mathcal{M} = \mathbb{R}^d$ and $d(x_i, x_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2$, MDS recovers PCA.
(Details: <https://mlatcl.github.io/advds/notes/mds.pdf>)

MDS Summary

MDS Summary

Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.		M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn

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MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn

When to use MDS:

MDS Summary

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LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn

When to use MDS:

- ▶ We have a distance matrix as input rather than a data matrix
- ▶ Preserve distances in reduced space
- ▶ Don't need an explicit DR mapping to run on test data
- ▶ Preserve the ambient space geometry in the reduced space

Isometric Mapping

Before Isomap... Nearest neighbors!

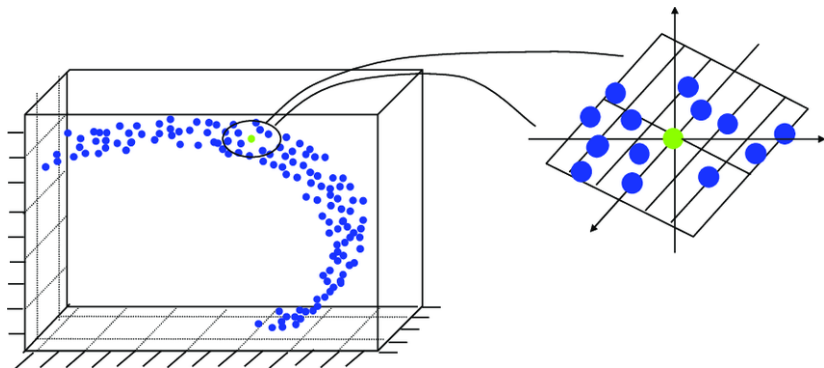
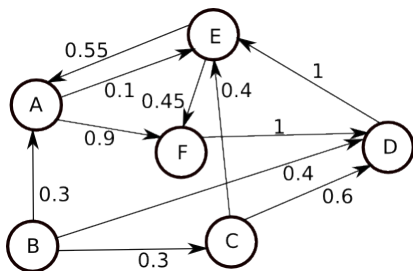


Image from: https://www.researchgate.net/publication/35199451_Location_Estimation_in_Sensor_Networks

Before Isomap... Graphs!

What is a graph?



A graph is collection of

- ▶ **Nodes** (e.g., *A*)
- ▶ **Edges** (e.g., arrow between *B* and *A*)
- ▶ **Edge weights** (e.g., 0.3)

Image from: <https://stackoverflow.com/questions/23947839/distance-between-two-nodes-in-weighted-directed-graph>

Nearest neighbor graph

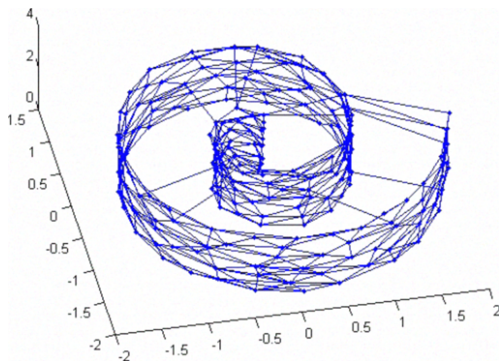


Image from: https://www.researchgate.net/publication/221531607_A_robust_nonlinear_projection_method_using_the_neural_as_network

What is Isomap?

Isomap (Isometric Mapping) is a nonlinear dimensionality reduction method that:

- ▶ Preserves **geodesic distances** (not just Euclidean distances)
- ▶ Extends classical MDS to nonlinear manifolds

Key idea: Approximate manifold distances with shortest paths on a neighborhood graph, then apply MDS.

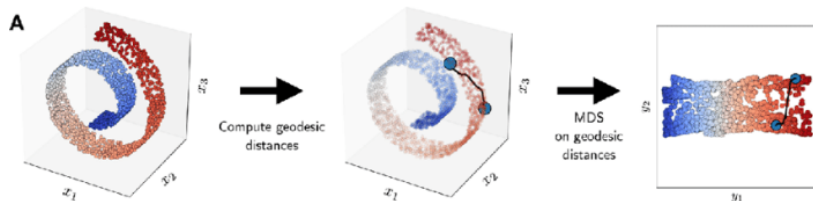


Figure: Figure borrowed from Tsai [2010].

Steps of Isomap

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Inputs: Distance, & number of nearest neighbors

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3. Apply classical MDS to D^{geo}

Isomap vs MDS

The advantage of a nearest-neighbors approach...

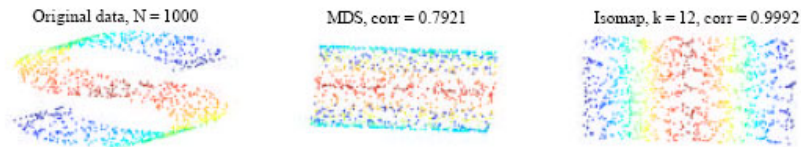


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LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn

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Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn

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- ▶ Your data lies on a low-dimensional **nonlinear** manifold

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- ▶ You want a global embedding that preserves intrinsic geometry

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Lab 5: MDS & Isomap

Go to `Lab5_IsomapMDS.ipynb`

t-SNE

Stochastic Neighbor Embedding (SNE): Overview

SNE constructs a probability distribution over neighbors for each point $\mathbf{x}_i$ based on distances in high-dimensional space.

$$g(\mathbf{x}_i, \mathbf{x}) = \exp \left(-\frac{\|\mathbf{x}_i - \mathbf{x}\|_2^2}{2\sigma_i^2} \right).$$

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The conditional probability that $\mathbf{x}_i$ picks $\mathbf{x}_j$ as neighbor:

$$p_{j|i} = \frac{g(\mathbf{x}_i, \mathbf{x}_j)}{\sum_{r \neq i} g(\mathbf{x}_i, \mathbf{x}_r)}.$$

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Perplexity and Bandwidth Selection

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Perplexity controls local scale for each point:

$$\text{Perp}(P_i) = 2^{H(P_i)}, \quad H(P_i) = - \sum_j p_{j|i} \log_2 p_{j|i}.$$

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- ▶ High perplexity $\implies$ broader neighborhoods

Binary search is used to find σ_i so that the perplexity matches a user-defined value.

- ▶ Small $\sigma_i \implies$ dense distribution
- ▶ Large $\sigma_i \implies$ sparse distribution

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Embed points $\mathbf{z}_i \in \mathbb{R}^k$ with kernel

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and conditional probabilities

$$q_{j|i} = \frac{h(\mathbf{z}_i, \mathbf{z}_j)}{\sum_{r \neq i} h(\mathbf{z}_i, \mathbf{z}_r)}.$$

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SNE uses gradient descent on $\{\mathbf{z}_i\}$, where gradients resemble forces pulling embedded points closer or pushing them apart.

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2. **Crowding problem:** Moderate distances in high-D collapse into tight clusters in low-D.
3. **Optimization difficulty:** KL divergence is sensitive & hard to optimize.

t-SNE: *Symmetrized* Probabilities

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t-SNE: *Symmetrized* Probabilities

t-SNE improves on SNE by:

- ▶ Using a **joint probability** distribution (with n samples)

$$p_{ij} = \frac{p_{i|j} + p_{j|i}}{2n}$$

- ▶ Modeling q_{ij} in the low-dimensional space with a **Student's-t distribution**:

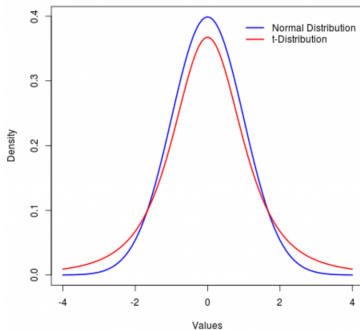
$$q_{ij} = \frac{(1 + \|\mathbf{z}_i - \mathbf{z}_j\|^2)^{-1}}{\sum_{r \neq s} (1 + \|\mathbf{z}_r - \mathbf{z}_s\|^2)^{-1}}$$

t-SNE: *Fixing the Crowding Problem*

Why use Student's-t?

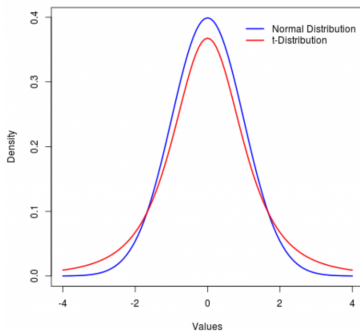
t-SNE: *Fixing the Crowding Problem*

Why use Student's-t?



t-SNE: *Fixing the Crowding Problem*

Why use Student's-t?



The heavy tails of the t-distribution allow moderate distances to be represented more accurately — solving the **crowding problem**.

t-SNE minimizes the KL divergence between p_{ij} and q_{ij} :

$$C = \sum_{i,j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right)$$

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Points repel or attract depending on similarity; repulsion is bounded (unlike in SNE) [Böhm et al., 2022]

Parameter Tuning

t-SNE parameters:

- ▶ **Perplexity** (typical range: 5–50)
- ▶ **Learning rate**
- ▶ **Number of iterations**

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Effect of tuning: t-SNE is sensitive to parameter choices... see this website for parameter tuning intuition

t-SNE objective function is non-convex, so run multiple times and keep the lowest objective value

Summary (Part 1)

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 - ▶ Better optimization
- ▶ t-SNE slow ($O(n^2)$), Barnes-Hut t-SNE is faster ($O(n(\log(n)))$)

Summary (Part 2)

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Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn

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MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn

When to use t-SNE

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When to use t-SNE

- ▶ t-SNE preserves local structure using probability distributions

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When to use t-SNE

- ▶ t-SNE preserves local structure using probability distributions
- ▶ Widely used for visualizing high-dimensional data

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Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn

When to use t-SNE

- ▶ t-SNE preserves local structure using probability distributions
- ▶ Widely used for visualizing high-dimensional data
- ▶ Often outputs *very* clustered data

Handbook for using t-SNE!

Lab 6: t-SNE

Go to Lab6_tSNE.ipynb &
`https://distill.pub/2016/misread-tsne/`

UMAP

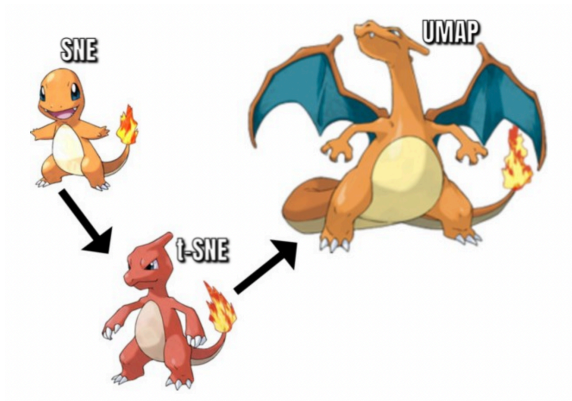


Figure: Image from <https://arize.com/blog-course/reduction-of-dimensionality-top-techniques/>.

Uniform Manifold Approximation (UMAP)

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Goal: preserve the topological structure of the data manifold

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Given dataset $\{\mathbf{x}_1, \dots, \mathbf{x}_p\}$

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User-specified:

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UMAP is based in category theory & mathematical topology.. we will sweep this foundation under the rug and consider the computational machinery instead

Algorithm Step 1: Ambient Graph Building

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3. Define edge weights: Edge weights from i to i_j :

$$\mathbf{w}_h(\mathbf{x}_i, \mathbf{x}_{i_j}) = \exp \left(\frac{-\max(0, d(\mathbf{x}_i, \mathbf{x}_{i_j}) - \rho_i)}{\sigma_i} \right)$$

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Interpretation: edge weights are probabilities that a 1-simplex exists;
combined weights: probability that at least one edge exists.

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Default parameters: $a = 1.577$, $b = 0.8951$

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$$C = \sum_e w_h(e) \log \left(\frac{w_h(e)}{w_l(e)} \right) + (1 - w_h(e)) \log \left(\frac{1 - w_h(e)}{1 - w_l(e)} \right)$$

Parameter Tuning & Resources

Key Parameters:

- ▶ **Number of neighbors:** balance local vs. global structure
- ▶ **Minimum distance:** desired separation of points in latent space

Visualizing parameter tuning and comparison to t-SNE

Links & Resources:

- ▶ UMAP: Uniform Manifold Approximation and Projection
- ▶ UMAP Documentation
- ▶ Understanding UMAP
- ▶ Inverse transform
- ▶ Aligned UMAP

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[Link for a tutorial on how to use UMAP.](#)

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***We can make implicit methods explicit** by optimizing for function parameters for embeddings instead of embedding coordinates!!! Damrich et al. [2022]*

Lab 7: UMAP

Go to Lab7_UMAP.ipynb &
<https://pair-code.github.io/understanding-umap/>

Autoencoders

1. Vanilla Autoencoders
2. Denoising Autoencoders
3. Variational Autoencoders
4. Convolutional Autoencoders
5. Graph Autoencoders

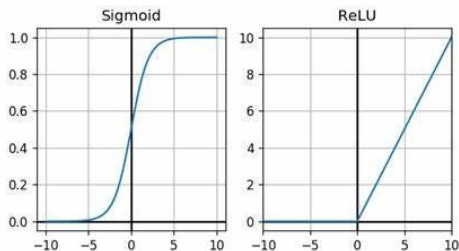
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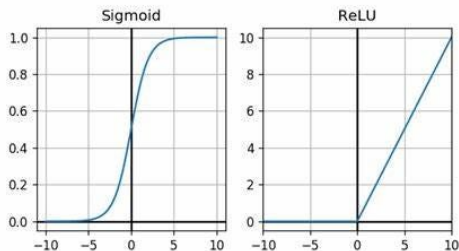
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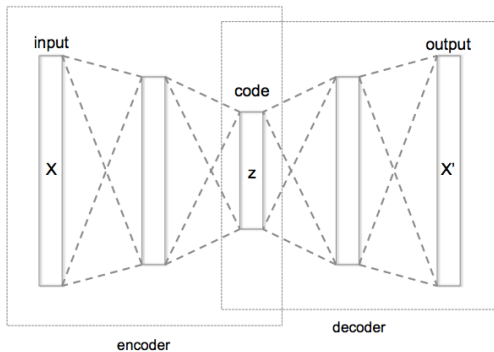


The “neural network function” maps

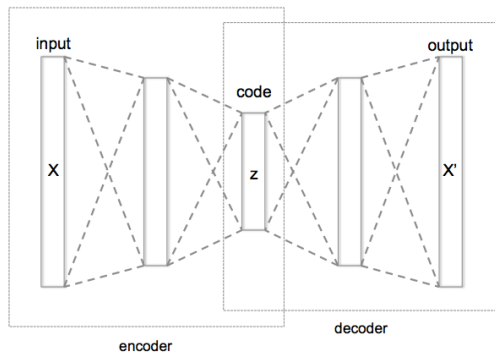
$$\mathbf{x} \mapsto \sigma(\mathbf{W}_\ell \sigma(\mathbf{W}_{\ell-1} \cdots \sigma(\mathbf{W}_1(\mathbf{x})) \cdots))$$

Vanilla Autoencoder

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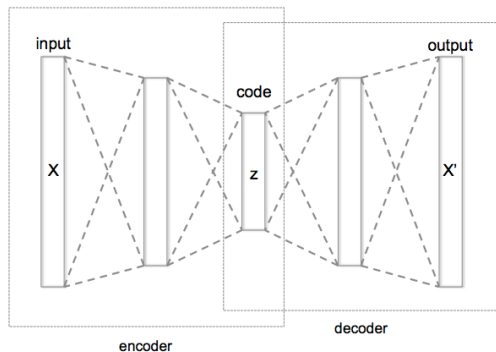


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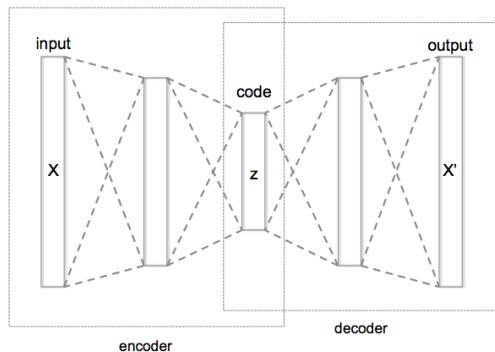
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Vanilla Autoencoder



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- ▶ Loss function: Typically Mean Squared Error (MSE) between the input and the reconstructed output

Autoencoders vs PCA

Autoencoders optimize reconstruction error

$$\arg \min_{\text{Enc}, \text{Dec}} \sum_{i=1}^n L(\mathbf{x}_i, \text{Dec} \circ \text{Enc}(\mathbf{x}_i)).$$

- ▶ Define $\text{Enc}(\mathbf{x}) = \mathbf{A}^\top \mathbf{x}$ and $\text{Dec}(\mathbf{z}) = \mathbf{A}\mathbf{z}$.
- ▶ Take the mean squared error (MSE) loss $L(\mathbf{x}, \hat{\mathbf{x}}) = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2$.

$$\arg \min_{\mathbf{A}^\top \mathbf{A} = \mathbf{I}} \sum_{i=1}^n \|\mathbf{x}_i - \mathbf{A}\mathbf{A}^\top \mathbf{x}_i\|_2^2.$$

- ▶ This is *exactly* PCA

But, there's a deeper connection... see Rolinek et al. [2019]; Bao et al. [2020].

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Problem: Marginal likelihood is intractable because we don't know the posterior

$$p_\theta(\mathbf{x}) = \int_{\mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int_{\mathbf{z}} p_\theta(\mathbf{z}) p_\theta(\mathbf{x}|\mathbf{z}) d\mathbf{z}$$

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Although $p_{\theta}(\mathbf{z}|\mathbf{x})$ is also intractable... we can estimate it!

Estimate $p_{\theta}(\mathbf{z}|\mathbf{x})$

Goal: approximate $p = p_{\theta}(\mathbf{z}|\mathbf{x})$ with $q = q_{\phi}(\mathbf{z}|\mathbf{x})$.

KL Divergence

$$D_{\text{KL}}(q||p) = \int_{\mathbf{z}} q \log(q/p) d\mathbf{z} \quad (7)$$

$$= \mathbb{E}_q [\log(q/p)] \quad (8)$$

$$= \mathbb{E}_q [\log q] - \mathbb{E}_q [\log p] \quad (9)$$

$$= \mathbb{E}_q [\log q] - \mathbb{E}_q [\log(p_{\theta}(\mathbf{z}, \mathbf{x})/p_{\theta}(\mathbf{x}))] \quad (10)$$

$$= \mathbb{E}_q [\log q] - \mathbb{E}_q [\log p_{\theta}(\mathbf{z}, \mathbf{x})] + \mathbb{E}_q [\log p_{\theta}(\mathbf{x})] \quad (11)$$

$$= \log p_{\theta}(\mathbf{x}) - \mathbb{E}_q [\log p_{\theta}(\mathbf{z}, \mathbf{x}) - \log q_{\phi}(\mathbf{z}|\mathbf{x})] \quad (12)$$

The expected information loss when we model p with q .

Problem: $\log p_{\theta}(\mathbf{x})$ intractible

More with KL Divergence

From before, we have

$$\log p_{\theta}(x) = D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p_{\theta}(\mathbf{z}|\mathbf{x})) + \mathbb{E}_q[\log p_{\theta}(\mathbf{z}, \mathbf{x}) - \log q_{\phi}(\mathbf{z}|\mathbf{x})]$$

Notice $D_{\text{KL}}(q||p) > 0$

Enter the Evidence Lower BOund (ELBO)!

$$\log p_{\theta}(x) \geq \mathbb{E}_q[\log p_{\theta}(\mathbf{z}, \mathbf{x}) - \log q_{\phi}(\mathbf{z}|\mathbf{x})] = \text{ELBO}$$

The Glory of ELBO

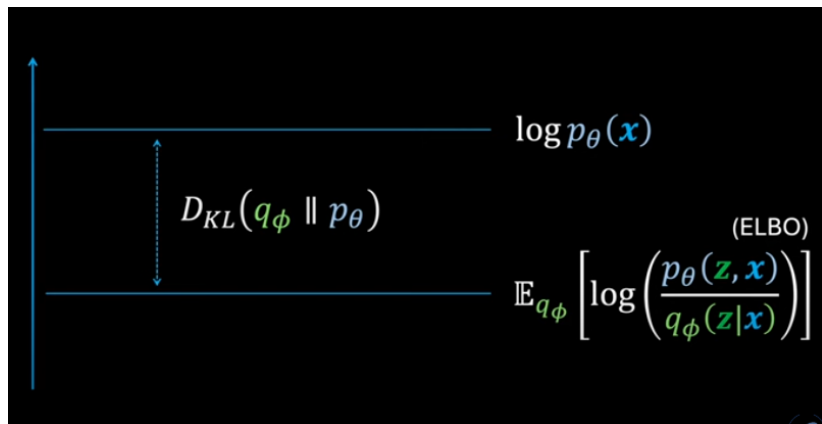


Figure: Image borrowed from
<https://www.youtube.com/watch?v=HBYQvK1aE0A>.

Maximizing ELBO maximizes $p_{\theta}(\mathbf{x})$ and $D_{KL}(q||p)$!

Stochastic gradient descent on ELBO... but issues with estimation of gradient...

Reparameterization Trick:

- ▶ $q_{\phi}(\mathbf{z}|\mathbf{x}) = g(\phi, \mathbf{x}, \epsilon)$
- ▶ $\epsilon \sim p(\epsilon)$

Now expectation is taken over $p(\epsilon)$, rather than q

$$\mathcal{L}(\mathbf{x}) = \mathbb{E}_{p(\epsilon)}[\log p_{\theta}(\mathbf{z}, \mathbf{x}) - \log q_{\phi}(\mathbf{z}|\mathbf{x})]$$

Exercise: we approximate ELBO with L samples using

$$\text{ELBO} \approx \frac{1}{L} \sum_{i=1}^L \log p_{\theta}(\mathbf{x}|\mathbf{z}) - \text{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) || p_{\theta}(\mathbf{z}))$$

Latent Space Simplifying Assumptions

From before ... the VAE is trained to maximize

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Gaussian Assumptions

- ▶ Prior: $p(\mathbf{z}) = \mathcal{N}(0, I)$
- ▶ Posterior: $q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma})$
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Exercise: Show that this loss becomes

$$\text{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p_{\theta}(\mathbf{z})) = \frac{-1}{2} [\log \boldsymbol{\sigma}^2 - \boldsymbol{\mu}^2 - \boldsymbol{\sigma}^2 + 1].$$

VAE Architecture

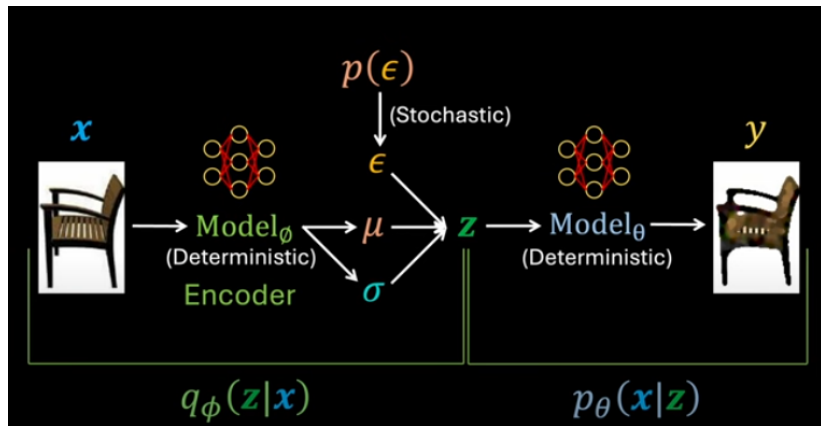


Figure: Image borrowed from <https://www.youtube.com/watch?v=HBYQvK1aEOA>.

The VAE Loss

$$-\text{ELBO} \approx -\frac{1}{L} \sum_{i=1}^L \log p_{\theta}(\mathbf{x}|\mathbf{z}) + \text{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p_{\theta}(\mathbf{z}))$$

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“Reconstruction error” $\log p_{\theta}(\mathbf{x}|\mathbf{z})$

- ▶ Gaussian assumption: $(\mathbf{x} - \hat{\mathbf{x}})^2$ (*takes work to show*)
- ▶ Bernoulli assumption: $\text{BCE}(\hat{\mathbf{x}}, \mathbf{x})$

Similarity between predicted $q_{\phi}(\mathbf{z}|\mathbf{x})$ & true $p_{\theta}(\mathbf{z})$ latent distributions

We assume $p_{\theta}(\mathbf{z})$ is Gaussian: $p_{\theta}(\mathbf{z}) = \mathcal{N}(0, I)$

When to use VAEs: generative modeling and denoising

VAE Summary

When to use VAEs: generative modeling and denoising

Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.		M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn
UMAP	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	github
DAE	✓	✗	✗	✗	✓	✓	✗	✗	✗	✗	github
VAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✗	github

Convolutional Autoencoder

Continuous Convolution

Continuous Convolution

Consider functions: $f, g: \mathbb{R} \rightarrow \mathbb{R}$.

$$f * g(x) = \int_u f(u)g(x - u)du$$

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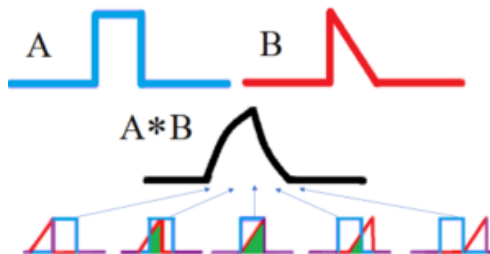


Figure:

<https://www.statisticshowto.com/convolution-integral-simple-definition/>

Discrete Convolution

Discrete Convolution

Consider functions: $f, g: \mathbb{Z} \rightarrow \mathbb{Z}$.

$$f * g(n) = \sum_m f(m)g(n - m)$$

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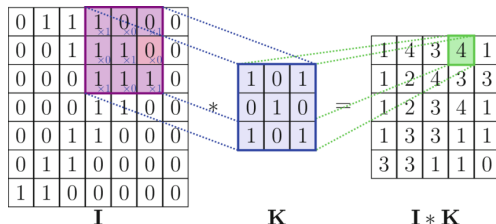


Figure: <https://www.researchgate.net/figure/>

A-visualization-of-the-discrete-convolution-operation-in-2D_fig5_372609466

Convolutional Filter

Sliding convolutional filter.

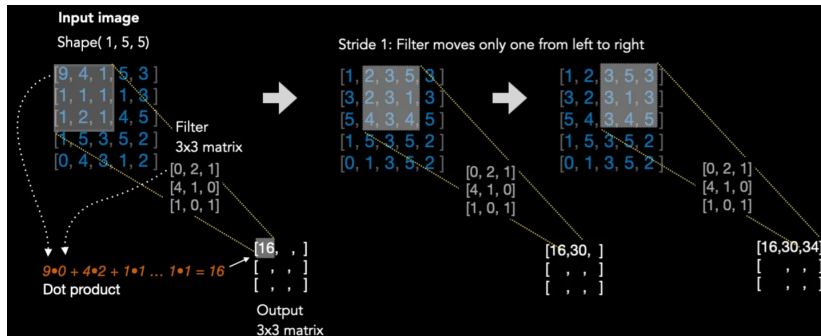


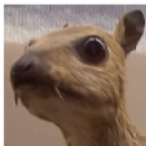
Figure:

[https://medium.com/advanced-deep-learning/cnn-operation-with-2-kernels
-resulting-in-2-feature-mapsunderstanding-the-convolutional-filter-c4aad26cf32](https://medium.com/advanced-deep-learning/cnn-operation-with-2-kernels-resulting-in-2-feature-mapsunderstanding-the-convolutional-filter-c4aad26cf32)

Edge Detectors

Certain filters extract different patterns.

Input image



Convolution
Kernel

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Feature map



Figure: <https://timdettmers.com/2015/03/26/convolution-deep-learning/>

Convolutional Autoencoders

Encoder and decoder are convolutional filters

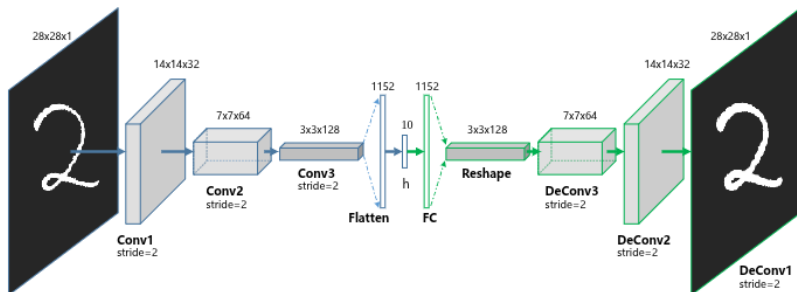


Figure: [Guo et al., 2017]

When to use CAEs: DR for image datasets

CAE Summary

When to use CAEs: DR for image datasets

Method	Map		Data Sup.	Constr. & Prop. Pres.							Code
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Graph Convolutional Autoencoder

Convolutions and Fourier Transform

Convolutions and Fourier Transform

Let $\mathcal{F}$ be the Fourier transform. Consider functions: $f, g: \mathbb{R} \rightarrow \mathbb{R}$. Then, under certain assumptions

$$\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$$

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$$\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$$

- ▶ Multiplication is easy!
- ▶ Decreases the computational cost of convolution

Discrete Fourier Transform (DFT)

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$$\mathbf{V} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & \omega & \cdots & \omega^{d-1} \\ 1 & \omega^2 & \cdots & \omega^{2(d-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{p-1} & \cdots & \omega^{(d-1)(d-1)} \end{bmatrix}.$$

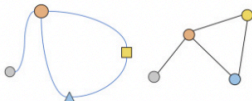
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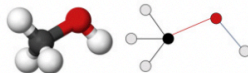
$$\mathbf{V} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & \omega & \cdots & \omega^{d-1} \\ 1 & \omega^2 & \cdots & \omega^{2(d-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{p-1} & \cdots & \omega^{(d-1)(d-1)} \end{bmatrix}.$$

- ▶ The DFT of a timeseries $\mathbf{x} = [x(1)|x(2)|\cdots|x(d)]^\top$ is $\mathbf{V}\mathbf{x}$
- ▶ Transforms from time to frequency domain
- ▶ Can also be used on other signals (features other than time)

Graph Data



(a) Physics



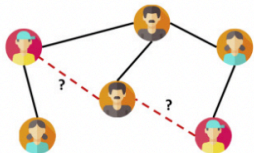
(b) Molecule



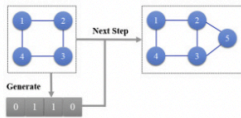
(c) Image



(d) Text



(e) Social Network

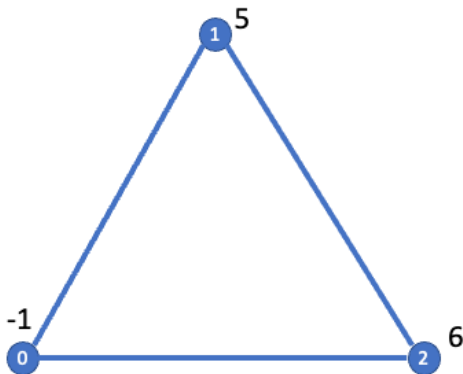


(f) Generation

Figure: [Zhou et al., 2020]

Graph Signals

A graph signal is a vector with a value at every node on the graph.



Discrete Graph Fourier Transform (GFT)

1. $\mathbf{A}$ adjacency matrix for the graph

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On the cycle graph, the GFT reduces to the DFT.

Graph Convolution

Graph signals $\mathbf{x}, \mathbf{g}$

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$$\mathcal{F}(\mathbf{x} * \mathbf{g}) = \mathcal{F}(\mathbf{x}) \cdot \mathcal{F}(\mathbf{g}) = \mathbf{\Lambda}_{\mathbf{U}^\top \mathbf{g}} \mathbf{U}^\top \mathbf{x}$$

Where $\cdot$ is point-wise multiplication and

$$\mathbf{\Lambda}_{\mathbf{g}} = \begin{bmatrix} (\mathbf{U}^\top \mathbf{g})_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & (\mathbf{U}^\top \mathbf{g})_d \end{bmatrix}$$

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Graph Convolutional Layer [Kipf and Welling, 2016a]

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Renormalization trick & gives us the graph convolutional layer:

$$\text{GC}(\mathbf{X}) = \sigma(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2} \mathbf{X} \Theta) \quad (13)$$

- ▶ $\tilde{\mathbf{A}} = \mathbf{I} + \mathbf{A}$
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See Kipf and Welling [2016b] for details.

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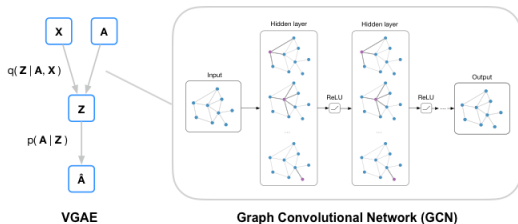


Figure: Code and image from <https://github.com/tkipf/gae>

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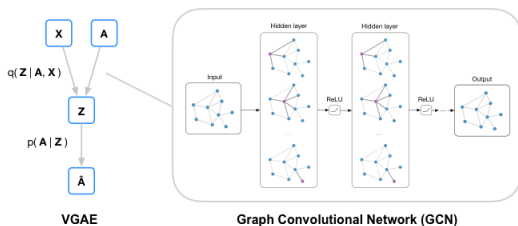


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Further reading: graph-based architectures [Wu et al., 2020] and generalizations of graph autoencoders [Hajij et al., 2021]

When to use VGAEs: generative modeling and graph-structured data.

VGAE Summary

When to use VGAEs: generative modeling and graph-structured data.

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VAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✗	github
VGAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✓	github

DR for Visualization

Notes from a CVPR Tutorial

Practical Tips for DR Visualization

Do's:

- ▶ Each DR method has a **scale parameter** (e.g., kernel bandwidth, perplexity) — explore its effect.
- ▶ Be creative with **inputs** (e.g., similarity, probability, or relational data).
- ▶ Be creative with **outputs** (e.g., use low-D coords as RGB values or overlays).
- ▶ Look for **patterns, clusters, and relations**.
- ▶ Use DR to **generate hypotheses**, not to prove them.
- ▶ Explore time-series visualization using DR is an emerging research direction.

Don'ts:

- ▶ Avoid “**proof by visualization.**” DR plots are for exploration, not evidence. Don't draw firm conclusions solely from DR plots.
- ▶ Low-dimensional metric spaces cannot represent all non-metric similarities (real-world data often violates the triangle inequality).



Summary

- ▶ *Linear algebra review*
- ▶ *Taxonomy of DR*
- ▶ *Linear DR: PCA & LDA*
- ▶ *Nonlinear DR: MDS, Isomap, t-SNE, & UMAP*
- ▶ *Autoencoders: DAE, VAE, CAE, VGAE*

The end, thank you!

Special thanks to Gustau Camps-Valls, Kai
Hendrik-Cors, Homer Durand, Vassilis
Sitokonstantinou & Tristan Williams

Extra Topics

1. Gradient Descent
2. Dynamic Mode Decomposition [Tu, 2013]
3. Koopman Mode Decomposition [Brunton et al., 2021]

Gradient Descent

Gradient Descent: Overview

- ▶ Gradient Descent is an optimization algorithm to minimize a function
- ▶ It iteratively moves in the direction of steepest descent (negative gradient)

Objective

Minimize a loss function $f(\theta)$, where:

- ▶ θ are the model parameters

Update Rule

Update Rule

At each iteration, the parameters are updated as:

$$\theta_{t+1} = \theta_t - \eta \nabla f(\theta_t)$$

- ▶ θ_t : Current parameters at iteration t
- ▶ η : Learning rate (step size)
- ▶ $\nabla f(\theta_t)$: Gradient of the loss function

Dynamic Mode Decomposition

Dynamical Systems

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Discrete-time: $\mathbf{x}(t + \tau) = f(\mathbf{x}(t))$ Continuous-time: $\frac{d\mathbf{x}}{dt} = g(\mathbf{x}(t))$

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$\Rightarrow$ Solutions evolve through powers of $\mathbf{A}$: eigenvalues/eigenvectors govern behavior.

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- ▶ $\omega_j \in \mathbb{C}$ **temporal characteristics** (continuous time eigenvalues)
- ▶ $b_j \in \mathbb{R}$ scalar loadings (a.k.a. amplitudes)

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DMD analyzes the stability of this system and generates spatial patterns in $\mathbb{R}^n$ (ambient space)

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Solve via variable projection method.

Further Reading

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- ▶ Multiverse of DMD [Colbrook, 2023]
- ▶ Physics-informed DMD Baddoo et al. [2023]
- ▶ Generalizing DMD: Modern Koopman theory [Brunton et al., 2021]

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Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.		M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn
UMAP	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	github
D & CAE	✓	✗	✗	✗	✓	✓	✗	✗	✗	✗	github
VAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✗	github
VGAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✓	github
DMD	✓	✓	✗	✓	✗	✓	✗	✗	✗	✗	github

Bonus lab!

Tutorial 1 here.

Koopman Mode Decomposition

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Problem: there's no linear map that goes from $\mathbf{x}(t)$ to $\mathbf{x}(t + \tau)$...

Koopman Mode Decomposition

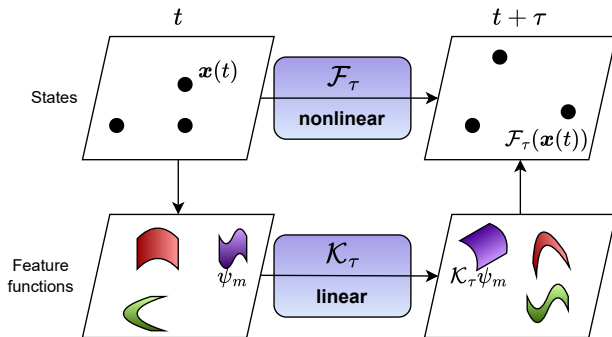
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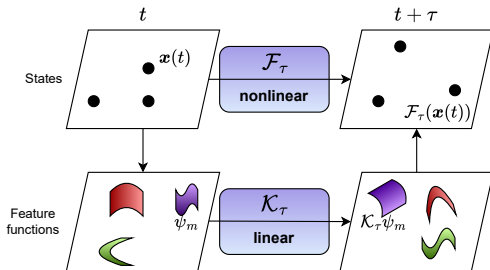
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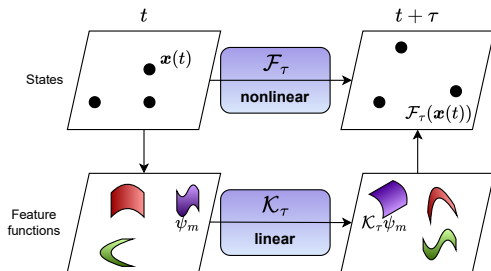
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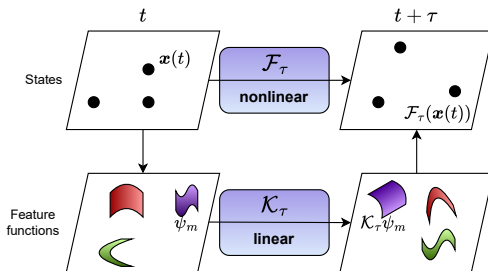
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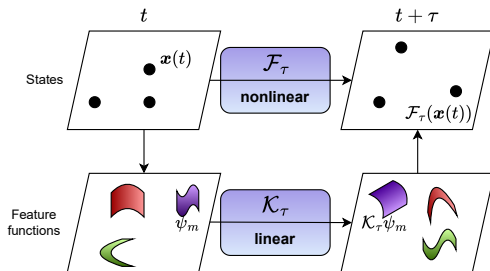
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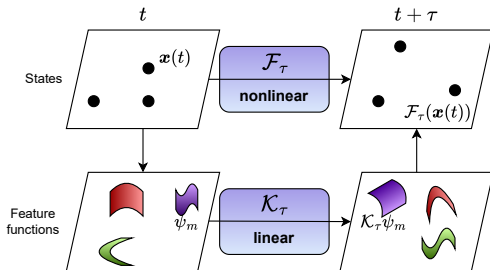
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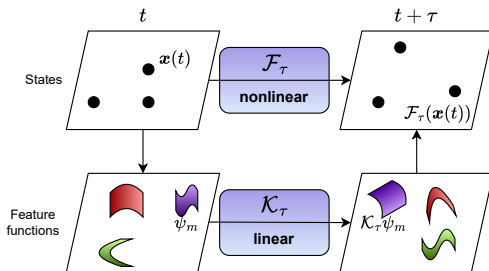
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- ▶ Define the snapshot feature matrices

$$\Psi_{\mathbf{x}} = \begin{bmatrix} \psi(\mathbf{x}(1)) & \cdots & \psi(\mathbf{x}(T - \tau)) \end{bmatrix} \quad (14)$$

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Kernel Koopman Mode Decomposition [Williams et al., 2015]

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$$\mathbf{K}_{\mathbf{x}} = \Psi_{\mathbf{x}}^{\top} \Psi_{\mathbf{x}}, \quad \mathbf{K}_{\mathbf{x}', \mathbf{x}} = \Psi_{\mathbf{x}'}^{\top} \Psi_{\mathbf{x}}.$$

Kernel Koopman Mode Decomposition

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Name	Definition
Eigenfunctions of $\mathbf{X}$	$\Phi = \mathbf{K}_X \mathbf{Q} \Sigma^\dagger \mathbf{W}$
Koopman modes	$\mathbf{B} = \mathbf{X} \mathbf{Q}^\top \Sigma^\dagger \mathbf{W}^{-1}$

KMD Summary

When to use KMD: nonlinear timeseries DR

Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.		M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn
UMAP	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	github
D & CAE	✓	✗	✗	✗	✓	✓	✗	✗	✗	✗	github
VAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✗	github
VGAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✓	github
DMD	✓	✓	✗	✓	✗	✓	✗	✗	✗	✗	github
KMD	✗	✓	✗	✓	✗	✓	✗	✗	✗	✗	github

Bonus lab!

Tutorials here:

- ▶ Extended (Kernel) DMD
- ▶ Koopman Mode Decomposition

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